Ringel, Graceful and Harmonious

Rainbow Trees in Colourings Defined by a Cyclic Group

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Recap from Decks 00 and 02

▷ $\mathbb{Z}_m$: integers mod m. Translation $x \mapsto x + g$ is a bijection of $\mathbb{Z}_m$ (deck 00).

▷ Deck 02: colour the arc $g \to gs$ of a Cayley digraph by s; "distinct partial products" became "rainbow path", and the proof used only that **translations preserve colours**.

▷ A copy of a graph H in an edge-coloured graph is **rainbow** if its edges have distinct colours.

Today: colour the edges of K_m by a *difference* $x - y$ or a *sum* $x + y$ in a group, and look for rainbow *trees*.

A Labelling Becomes an Embedding

Put the available labels on the vertices of a complete graph. Assigning distinct labels to $V(T)$ places a copy of T in that graph.

labelling condition	**condition on the embedded tree**
distinct absolute differences	rainbow in colour $\lvert x-y\rvert$
distinct cyclic lengths	rainbow in near-distance colours
distinct sums in a group	rainbow in colour $x+y$

The host and its label set matter: using $\{0,\dots,m\}$ is more restrictive than choosing $m+1$ vertices in $\mathbb{Z}_{2m+1}$ (MPS, §1). Harmonious trees allow one repeated vertex label; their rainbow reformulation will therefore require a separate implication (Müyesser–Pokrovskiy, §1).

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Decompositions, from Euler to Walecki

A graph G **decomposes** into copies of H if $E(G)$ is a disjoint union of edge sets of subgraphs isomorphic to H (Montgomery–Pokrovskiy–Sudakov, §1).

▷ Euler (1782): a Latin square into n transversals $\Leftrightarrow$ $K_{n,n,n,n}$ into copies of K_4 (deck 01).

▷ Kirkman (1847): K_n into triangles $\iff$ $n \equiv 1, 3 \pmod 6$.

▷ Steiner (1853), Keevash (2014): the existence of designs — decompositions into **small** cliques.

▷ Walecki (1882): K_{2n} into n Hamilton paths — the round dances of deck 02, a decomposition into **large** graphs.

Which other large graphs decompose a complete graph?

Ringel's Conjecture, 1963

Conjecture 1 (Ringel, 1963)

For every tree T with n edges, K_{2n+1} decomposes into $2n+1$ copies of T.

The numbers fit. K_{2n+1} has $\binom{2n+1}{2} = (2n+1)\,n$ edges: exactly $2n+1$ copies of an n-edge tree. Ringel asks whether this host works for every such tree; the edge count alone does not guarantee a decomposition.

Example 1 (two easy trees, on $\mathbb{Z}_{2n+1}$)

Stars: the star at i with leaves $i+1, \ldots, i+n$; the $2n+1$ stars use each edge $\{x, x+d\}$, $1 \le d \le n$, once.
Paths: Walecki's zigzag (deck 02) rotated $2n+1$ times (Lucas, 1883).

Rosa's Idea: Number the Vertices, 1966/67

Definition 2 (Rosa, 1967: β-valuation; "graceful" by Golomb)

A labelling $f\colon V(T) \to \{0, 1, \ldots, n\}$ of a tree with n edges is **graceful** if it is injective and the edge labels $|f(x) - f(y)|$ are distinct — hence exactly $1, 2, \ldots, n$.

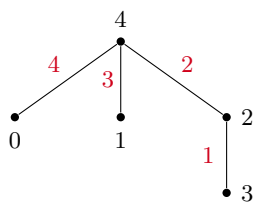


Figure: A graceful labelling of a tree with 4 edges; red: the edge labels $4, 3, 2, 1$.

Example 3 (path with 3 edges)

Labels $0, 3, 1, 2$ along the path: edge labels $3, 2, 1$.

Rosa introduced such labellings "as tools for decomposing the complete graph into isomorphic subgraphs" (Gallian's survey, §1).

Rosa's Theorem: Graceful Trees Decompose K_{2n+1}

Theorem 4 (Rosa, 1967; see MPS, §1)

If a tree T with n edges has a graceful labelling, then K_{2n+1} decomposes into $2n+1$ copies of T, all obtained from one by rotation.

Proof. Put T on the vertices $f(V(T)) \subseteq \{0, \dots, n\}$ of K_{2n+1} (vertex set $\mathbb{Z}_{2n+1}$), and let $T_i = T + i$ for $i \in \mathbb{Z}_{2n+1}$. The edge of T with label d joins some u and $u+d$. Every edge of K_{2n+1} has the form $\{x, x+d\}$ with a unique $d \in \{1, \dots, n\}$, and it lies in exactly one T_i, namely $i = x - u$. □

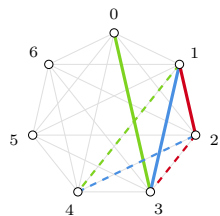


Figure: $n = 3$: the path $0, 3, 1, 2$ (solid) and its rotation by 1 (dashed); the 7 rotations decompose K_7 (checked).

The Graceful Tree Conjecture

Conjecture 2 (Rosa, 1966/67; also called the Ringel–Kotzig conjecture)

Every tree has a graceful labelling.

▷ It would give Ringel's conjecture, with a **cyclic** decomposition.

▷ Known for caterpillars (Rosa), trees with at most 4 leaves, diameter at most 5, symmetrical trees, olive trees, ...; all trees with at most 35 vertices, by computer (Fang, 2010) (Gallian, §2.1).

▷ Kotzig called the effort to prove it a "disease" (Gallian, §2.1).

▷ Rosa: graphs fail to be graceful for three reasons — too many vertices, too many edges, wrong parity (e.g. the cycles C_{4k+1}, C_{4k+2}) (Gallian, §1).

Rosa's ρ-Labellings and Kotzig's Conjecture

Graceful asks for labels in $\{0, \ldots, n\}$. For the decomposition only the differences *modulo* $2n+1$ matter.

Definition 5 (ρ-labelling, in modern form (MPS, §1))

An injective $f\colon V(T) \to \mathbb{Z}_{2n+1}$ such that the n edges have distinct **lengths** $\ell(f(x), f(y)) \in \{1, \ldots, n\}$, where $\ell(x, y) = \min(\, x - y,\ \ y - x\,)$ with $x - y,\ y - x$ read in $\{0, \ldots, 2n\}$.

Conjecture 3 (Kotzig; see Rosa, 1967 and MPS, §1)

*For every tree T with n edges, K_{2n+1} has a **cyclic** decomposition into copies of T: one invariant under $x \mapsto x+1$.*

Every graceful labelling is a ρ-labelling; the proof of Rosa's theorem uses only the ρ-property.

Graham and Sloane, 1980: Additive Bases

Problem 6 (Graham–Sloane, 1980, §1)

*$n_v(k)$: the largest n such that some k-set $A \subseteq \mathbb{Z}_n$ represents **every** element of $\mathbb{Z}_n$ as $a_i + a_j$ with $i < j$. $v_v(k)$: the smallest v such that some k-set $A \subseteq \mathbb{Z}_v$ has all sums $a_i + a_j$ $(i < j)$ **distinct**.*

Example 7 (their abstract)

$A = \{0, 1, 2, 4, 8, 13, 18, 22\}$: the 28 sums $a_i + a_j$ cover $\mathbb{Z}_{24}$, so $n_v(8) \geq 24$ (in fact $= 24$; covering checked).

Why. "Our interest in v_v stems from its application to **error-correcting codes**": sets with distinct sums mod v give large constant-weight binary codes $A(k, 6, w)$ (§1).

Harmonious Graphs

Definition 8 (Graham–Sloane, 1980, §3)

A connected graph with e edges is **harmonious** if its vertices can be labelled by distinct elements of $\mathbb{Z}_e$ so that the edge sums $f(x) + f(y)$ are distinct. For a **tree** ($e = |V| - 1$) exactly one label is repeated.

▷ Then the edge sums are exactly $0, 1, \ldots, e-1$.

▷ A harmonious labelling of K_v is exactly a set attaining $n_v(v) = v_v(v) = \binom{v}{2}$; hence K_v **is harmonious iff** $v \leq 4$ (Thm 7) — e.g. $\{0, 1, 2, 4\}$ for K_4 (checked).

▷ Almost all graphs are not harmonious; odd cycles, ladders, wheels are (abstract).

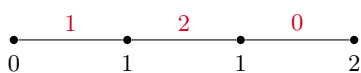


Figure: A path with 3 edges, labelled in $\mathbb{Z}_3$ (the label 1 repeated): edge sums $1, 2, 0$.

Harmonious Trees

Theorem 9 (Graham–Sloane, Thms 12–13)

Every caterpillar is harmonious. Every tree with at most 9 vertices is harmonious.

Conjecture 4 (Graham–Sloane, 1980, §5)

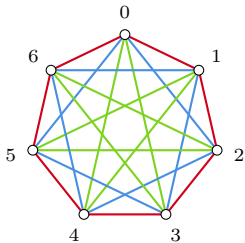
Every tree is harmonious.

▷ Verified for all trees with at most 31 vertices (Fang, 2011) (Gallian, §2.1).

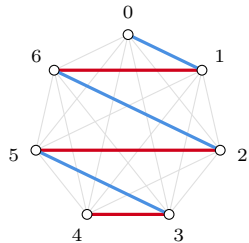
▷ Graham and Sloane write "cf. [Bloom]": the conjecture is the harmonious twin of the graceful one.

▷ Example: the fork of the previous slides, labelled $0, 1, 2, 1, 3$ in $\mathbb{Z}_4$ (checked).

A Picture: Chords of a Regular Polygon



colour = length:
1, 2, 3



colour = direction:
$x + y \equiv 0$, $x + y \equiv 1$

Figure: Vertices of a regular m-gon labelled $0, \ldots, m-1$. The chord $\{x, y\}$ has the same **length** as $\{x', y'\}$ iff $x - y \equiv \pm(x' - y')$, and is **parallel** to it iff $x + y \equiv x' + y'$ (Graham–Sloane, §3).

ρ-labelling: no two edges of equal length. Harmonious: no two edges parallel.

From Graphs to Groups: Harmonious Groups

Definition 10 (Beals–Gallian–Headley–Jungreis, 1991)

A finite group G is **harmonious** if its elements can be listed $g_1, \ldots, g_n$ so that $g_1g_2,\ g_2g_3,\ \ldots,\ g_{n-1}g_n,\ g_ng_1$ is again a list of all of G.

"Our definition was motivated by the concept of a harmonious graph invented by Graham and Sloane" (§1).

Fact 11 (Beals et al., §2)

*G is harmonious $\iff$ it has a **complete mapping** that is a single $|G|$-cycle (deck 01):* $g_1 \mapsto g_2 \mapsto \cdots \mapsto g_n \mapsto g_1$.

Example 12 (checked)

$\mathbb{Z}_7$: $0, 1, \ldots, 6$ (sums $1, 3, 5, 0, 2, 4, 6$). $\mathbb{Z}_2 \times \mathbb{Z}_4$: yes. $\mathbb{Z}_4$, $\mathbb{Z}_2^2$: no.

Two Algebraic Obstructions

Theorem 13 (Paige; Beals et al., Thm 3.1, Cor. 3.2)

If $g_1, \ldots, g_n$ is harmonious then $g_1 g_2 \cdots g_n$ lies in the commutator subgroup. Hence a group of even order with a ***cyclic Sylow 2-subgroup*** *is not harmonious — the Hall–Paige obstruction of deck 01.*

Theorem 14 (Beals et al., Thm 3.3)

Elementary abelian 2-groups $\mathbb{Z}_2^k$ are not harmonious: $gh = e$ forces $g = h$, so the identity is never a product of two consecutive (distinct) terms.

Theorem 15 (Beals et al., Thm 6.6)

A finite abelian group is harmonious $\iff$ its Sylow 2-subgroup is trivial or non-cyclic, and it is not elementary abelian of exponent 2.

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Hahn's Question, 1980

Problem 16 (Hahn, 1980, as quoted by Maamoun–Meyniel)

Let $n \geq 2k$, $k \geq 2$, $(n,k) \neq (4,2)$. In every colouring of the edges of K_n in which each colour appears ***at most k times****, is there a* ***rainbow Hamilton path****?*

Intuition: few edges per colour should make a rainbow path easy to find. A colouring is **proper** if adjacent edges get different colours; then each colour class is a matching, of size at most $n/2$.

An algebraic colouring answers it in the negative.

Maamoun and Meyniel's Answer, 1984

Theorem 17 (Maamoun–Meyniel, 1984)

Let $n = 2^p$, $p \geq 2$, identify $V(K_n)$ with $\mathbb{F}_2^p$ and give xy the colour $x + y$. This colouring is proper, every colour appears exactly $n/2$ times, and there is ***no rainbow Hamilton path****.*

Proof. A rainbow Hamilton path $x_1, \ldots, x_n$ has $n - 1$ edges, so it uses each non-zero colour once. In $\mathbb{F}_2^p$,

$$x_n - x_1 = \sum_i (x_{i+1} - x_i) = \sum_{c \neq 0} c = 0,$$

so $x_n = x_1$, a contradiction. □

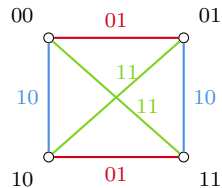


Figure: $p = 2$: each colour is a perfect matching; none of the 24 orderings of the vertices is rainbow (checked).

Buratti's Problem, 2007

For $x, y \in \mathbb{Z}_v$ the **length** of the edge xy of K_v is $\ell(x, y) = \min(|x - y|,\ v - |x - y|)$ (Horak–Rosa, §1).

Conjecture 5 (Buratti, 2007, communicated to Rosa)

Let $p = 2n + 1$ be prime and L any multiset of $2n$ elements of $\{1, \ldots, n\}$. Then K_p has a Hamilton path whose multiset of edge lengths is exactly L.

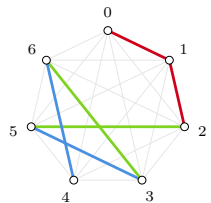


Figure: $p = 7$, $L = \{1, 1, 2, 2, 3, 3\}$: the path $0, 1, 2, 5, 3, 6, 4$.

▷ One length only: easy. Two lengths: solved (Horak–Rosa, 2009).

▷ All $p \leq 23$ by computer (Meszka) (Horak–Rosa, §1); for $p = 7$ all 28 lists are realised (checked).

Horak and Rosa: Any v, and a Divisor Condition

Conjecture 6 (Horak–Rosa, 2009; in the form of Pasotti–Pellegrini)

A multiset L of $v-1$ elements of $\{1,\dots,\lfloor v/2\rfloor\}$ is the list of edge lengths of a Hamilton path of K_v $\iff$ for every divisor d of v, at most $v-d$ elements of L are multiples of d.

Why the condition is necessary. An edge whose length is a multiple of d joins two vertices in the same coset of the subgroup $d\mathbb{Z}_v$, which has d cosets. These edges of the path form a forest whose components stay inside cosets, so it has at least d components, hence at most $v-d$ edges.

Example 18

$v=4$, $L=\{2,2,2\}$: three multiples of 2 but $v-d=2$. Indeed only the two edges $\{0,2\},\{1,3\}$ have length 2 (checked).

For $v=p$ prime the condition is empty: this is Buratti's conjecture.

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Three Colourings of Complete Graphs

Definition 19

1. The **near-distance (ND) colouring** of K_{2n+1} on $\mathbb{Z}_{2n+1}$: xy gets $\ell(x, y) \in \{1, \ldots, n\}$.
2. The **graceful colouring** of K_{n+1} on $\{0, \ldots, n\}$: xy gets $|x - y|$.
3. For an abelian group G, the **sum colouring** K_G: xy gets $x + y$ (Müyesser–Pokrovskiy, Def. 1.2).

Lemma 20

(a) In the ND colouring each colour class is a 2-regular spanning graph, and every translation $x \mapsto x + g$ preserves all colours. (b) In K_G each colour class is a matching, and $x \mapsto x + g$ maps colour c to colour $c + 2g$.

(a) $\ell(x + g, y + g) = \ell(x, y)$; (b) $(x + g) + (y + g) = c + 2g$.

Proposition 1: Kotzig's Translation

Proposition 21

*Let T be a tree with n edges. K_{2n+1} has a cyclic decomposition into copies of T $\iff$ the ND-coloured K_{2n+1} contains a **rainbow** copy of T $\iff$ T has a ρ-labelling.*

Proof. A rainbow copy is the same as a ρ-labelling. Rainbow $\Rightarrow$ cyclic decomposition: the argument of Rosa's theorem (the $2n+1$ translates; Lemma (a)).
Thus a rainbow tree supplies the complete decomposition by translation. The reverse direction requires checking that all copies form one orbit.

Why a Cyclic Decomposition Gives a Rainbow Tree

Let T have n edges, and suppose a decomposition of K_{2n+1} into copies of T is invariant under translation in $\mathbb{Z}_{2n+1}$.

1. No nonzero translation can fix a copy. Otherwise it induces a fixed-point-free automorphism of odd order on a tree. A tree automorphism fixes a vertex or an edge, and an odd-order map cannot swap an edge's ends.
2. Each orbit of copies therefore has size $2n+1$. There are exactly $2n+1$ copies, so they form one orbit $T_0 + i$.
3. Each length class has $2n+1$ edges. Translation preserves lengths, so T_0 contains exactly one edge of each length.

Hence T_0 is rainbow in the near-distance colouring, completing the equivalence with a ρ-labelling (the cyclic formulation, §1).

Proposition 2: Graceful Labellings

Proposition 22

For a tree T with n edges, the following are equivalent:

1. *T has a graceful labelling;*
2. *the graceful colouring of K_{n+1} has a rainbow **spanning** copy of T;*
3. *the ND-coloured K_{2n+1} has a rainbow copy of T whose vertex set is $\{0, 1, \ldots, n\}$.*

Proof. (1)⇔(2): a bijection $V(T) \to \{0, \ldots, n\}$ is a spanning copy; "distinct $|f(x) - f(y)|$" is "rainbow". (2)⇔(3): for $x, y \in \{0, \ldots, n\}$, $|x - y| \leq n$, so $\ell(x, y) = |x - y|$. □

Fixing the vertex set "adds an additional restriction which makes it harder to find such a rainbow copy" (MPS, §1).

Proposition 3: Harmonious Labellings

Proposition 23 (Müyesser–Pokrovskiy, §1)

Let T have n edges and let $T' = T - v$ for a leaf v. If $K_{\mathbb{Z}_n}$ contains a rainbow copy of T' (on all n vertices), then T is harmonious.

Proof. The copy $\varphi : V(T') \to \mathbb{Z}_n$ is a bijection with $n-1$ distinct edge sums, so exactly one colour c is unused. Let w be the neighbour of v and put $\varphi(v) = c - \varphi(w)$. Now the n edge sums are all of $\mathbb{Z}_n$, and exactly one label is repeated. □

The converse can fail (the repeated label need not sit on a leaf), so the rainbow statement is **stronger** than harmonious.

Proposition 4: Harmonious Groups as Rainbow Cycles

Proposition 24 (Beals et al., §2)

*Colour the arc $x \to y$ ($x \neq y$) of the complete digraph on G by xy. Then G is harmonious $\iff$ this digraph has a **rainbow directed Hamilton cycle**. For abelian G: $\iff$ K_G has a **rainbow Hamilton cycle**.*

Proof. A harmonious list $g_1, \ldots, g_n$ is the Hamilton cycle $g_1 \to g_2 \to \cdots \to g_n \to g_1$, whose n arc colours $g_i g_{i+1}$ are all of G, hence distinct; conversely a rainbow Hamilton cycle has n distinct colours from a set of size n. For abelian G, $xy = yx$, so arcs can be replaced by edges. □

Compare deck 02: there $x \to xs$ is coloured s (a *difference*); here $x \to y$ is coloured xy (a *product*).

Proposition 5: Buratti's Problem as Partial Sums

Proposition 25 (Ollis–Pasotti–Pellegrini–Schmitt, §1)

For a multiset L of $v-1$ elements of $\{1,\dots,\lfloor v/2\rfloor\}$ the following are equivalent:

1. *K_v has a Hamilton path with edge lengths L;*
2. *$L=\{a_1,\dots,a_{v-1}\}$ can be ordered and given signs $\varepsilon_i=\pm1$ so that the partial sums of $\varepsilon_1 a_1,\dots,\varepsilon_{v-1}a_{v-1}$ are exactly $\mathbb{Z}_v\setminus\{0\}$;*
3. *the circulant multigraph $\mathrm{Cay}[\mathbb{Z}_v : L\cup -L]$ has a cyclic decomposition into Hamilton paths (Pasotti–Pellegrini).*

Proof of (1)⇔(2). Translate the path to start at 0. Its steps $x_{i+1}-x_i=\pm a_i$ have lengths a_i, and its vertices $0,x_1,\dots,x_{v-1}$ are the partial sums; "Hamilton" means they are all of $\mathbb{Z}_v$. □

Example ($p=7$): steps $1,1,3,-2,3,-2$, partial sums $1,2,5,3,6,4$.

The Dictionary

labelling / group	**rainbow subgraph**
ρ-labelling of T	rainbow T in ND-coloured K_{2n+1}
graceful labelling of T	rainbow spanning T, graceful colouring of K_{n+1}
harmonious labelling of T	$\Leftarrow$ rainbow $T - v$ in $K_{\mathbb{Z}_n}$
harmonious group G	rainbow directed Hamilton cycle, arc $x \to y$ coloured xy
Buratti / Horak–Rosa	Hamilton path with prescribed colour multiset
cyclic decomposition of K_{2n+1}	one rainbow copy, translated
Maamoun–Meyniel	no rainbow Hamilton path in $K_{\mathbb{F}_2^p}$

Differences colour by length; sums colour by direction.

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Ringel's and Kotzig's Conjectures

▷ Ringel's conjecture for very large bounded-degree trees (Joos–Kim–Kühn–Osthus); approximate versions for $\Delta(T) = O(n/\log n)$ (Ferber–Samotij; Adamaszek–Allen–Grosu–Hladký) (MPS, §1).

▷ (Montgomery–Pokrovskiy–Sudakov, 2018) every 2-factorised K_{2n+1} contains a rainbow copy of every tree with $(1 - o(1))n$ edges.

Theorem 26 (Montgomery–Pokrovskiy–Sudakov; Keevash–Staden, 2020)

For all sufficiently large n, the ND-coloured K_{2n+1} contains a rainbow copy of every tree with n edges. Hence Ringel's and Kotzig's conjectures hold for large n.

Methods: absorption; randomised embeddings; a deterministic, almost graceful embedding for trees with very high degrees (MPS, §1).

Returning to the Fixed Label Interval

Ringel–Kotzig allows labels throughout $\mathbb{Z}_{2m+1}$ for an m-edge tree. Gracefulness still requires the fixed interval $\{0, \dots, m\}$.
From here, write $n = |V(T)|$, so the exact target is $n - 1$ distinct differences.

Theorem 27 (Adamaszek–Allen–Grosu–Hladký)

For every $\varepsilon > 0$, there are $\eta > 0$ and n_0 such that, if $n \geq n_0$ and $\Delta(T) \leq \eta n / \log n$, there is an injective labelling into $\{1, \dots, \lceil (1+\varepsilon)n \rceil\}$ with all differences distinct.

This retains every edge colour but enlarges the label interval. The next result keeps exactly n labels and counts how many colours survive.

Gracesize in the Difference-Coloured Complete Graph

Definition 28 (Letzter–Pokrovskiy–Williams, §1)

The gracesize $\mathrm{gs}(T)$ is the largest number of distinct edge differences under a bijection $V(T) \to \{1, \dots, n\}$.

Colour ij in $K_{\{1,\dots,n\}}$ by $|i-j|$. Then

$$\mathrm{gs}(T) = \max_{\text{spanning copies } T' \cong T} |\{\text{colours on } E(T')\}|.$$

Example 29 (same path, different vertex orders)

On four labels, the path $1, 2, 3, 4$ has differences $1, 1, 1$. The path $1, 4, 2, 3$ has differences $3, 2, 1$ and is graceful.

Exact gracefulness means one spanning copy uses all $n-1$ colours.

The 2026 Gracesize Bound: Every Large Tree

Theorem 30 (Letzter–Pokrovskiy–Williams, 2025; revised July 2026)

For every $\varepsilon > 0$ and sufficiently large n, every n-vertex tree has a bijective labelling by $\{1, \dots, n\}$ with at least $(1-\varepsilon)n$ distinct edge differences.

▷ In the difference colouring, some spanning copy of T uses at least $(1-\varepsilon)n$ colours. There is **no maximum-degree restriction**.

▷ This counts colours on the whole tree. Selecting one edge of each colour need not leave a connected subtree.

The next results ask for **all $n-1$ colours**, but restrict the shape of the tree.

Graceful Spiders: Widely Separated Leg Lengths

A spider consists of legs meeting at one centre; write ℓ_i for the number of edges in leg i.

Theorem 31 (Shan–Zhong, May 2026, Thm. 3)

A spider with $s \geq 2$ legs is graceful if

$$\ell_{i+1} \geq 2\ell_i + 2 \quad (2 \leq i < s), \qquad \ell_2 \geq \begin{cases} 2\ell_1 + 2, & \ell_2 \not\equiv 1 \pmod 4, \\ 2\ell_1 + 4, & \ell_2 \equiv 1 \pmod 4. \end{cases}$$

The leg lengths grow fast enough for this exact graceful labelling. In the difference-coloured $K_{|V(T)|}$, the spanning spider is rainbow.

Graceful Spiders with Few Long Legs

A different family allows arbitrarily many short legs.

Theorem 32 (Shan–Zhong, May 2026, Thm. 4)

A spider with one leg of arbitrary length and all other legs of length at most 2 has a graceful labelling with centre label 0.

Theorem 33 (Shan–Zhong, Thm. 5)

Every spider with at most three legs longer than 2 is graceful.

The first conclusion also **prescribes the centre label**. In the rainbow embedding, that specifies which host vertex is the centre. The second theorem only asserts gracefulness.

Graceful Combs with Prescribed Spine Labels

Theorem 34 (Dumitru–Nacu, July 2026, Thm. 2.4)

Every comb full binary tree on n vertices has a graceful labelling with alternating extremes $0, n-1, 1, n-2, \ldots$ pinned along a longest root-to-leaf spine.

A full binary tree has two children at each internal vertex; in a comb, the internal vertices lie on one root-to-leaf path. Combs are already caterpillars. The new conclusion prescribes the spine labels, so the rainbow embedding has additional fixed vertex positions.

Graceful Spiders with Many Added Leaves

We now fix the long legs and allow the number of leaves at the hub to grow.

Theorem 35 (Dumitru–Nacu, July 2026, Cor. 3.6)

Fix leg lengths $c, L_1, \ldots, L_r \geq 2$. The spider $S(c, L_1, \ldots, L_r, 1^t)$ has a graceful labelling with hub label 1 for all sufficiently large t.

▷ 1^t means t additional pendant edges at the common centre.

▷ For each fixed collection of long legs, adding enough such edges gives a rainbow spanning spider in the difference colouring.

▷ The hub occupies host vertex 1. The threshold may depend on the fixed long-leg lengths.

From Differences Back to Sums

We now change the edge colour from $|x-y|$ to $x+y$ in an abelian group G. A rainbow spanning tree is an injective labelling with distinct edge sums.

Theorem 36 (Müyesser–Pokrovskiy, 2025, revised October 2025)

Fix Δ. For large n, an n-vertex tree T with $\Delta(T) \leq \Delta$ has a rainbow copy in K_G, $|G| = n$, unless one of the three obstructions on the next slide occurs.

The leaf-extension implication proved earlier is the link to harmonious trees: embed $T - v$ with distinct sums, then use the missing sum to label v.

The paper settles the Graham–Sloane and Chang–Hsu–Rogers conjectures for bounded-degree trees. This is the **2025 baseline** before the new group-ordering results.

The Three Obstructions in the Sum Colouring

In the preceding theorem, G is abelian of order n and T has n vertices. Its exponent m is the least positive integer with $mg = 0$ for all $g \in G$.

Exceptional pairs (G, T) (Müyesser–Pokrovskiy)

1. $G = \mathbb{Z}_2^k$, and T is a path or has exactly two even-degree vertices;
2. G has exponent m, adjacent u, v have degrees $0 \pmod m$, and all other vertices have degrees $1 \pmod m$;
3. $G = \mathbb{Z}_2^k$, and the four even-degree vertices of T carry a perfect matching of T.

These are conditions on both the group and the degree pattern of the tree. They explain why a theorem for arbitrary sum colourings needs exceptions.

Harmonious Groups: From Existence to Symmetry

Return to cyclic orderings of a whole group: vertices and adjacent products must each enumerate G. The graph object is now a rainbow Hamilton cycle.

▷ Abelian groups are harmonious except elementary abelian 2-groups and groups with a unique involution; every odd-order group is harmonious (Beals et al., 1991).

▷ Every sufficiently large group satisfying the Hall–Paige condition, other than $\mathbb{Z}_2^k$, is harmonious (Müyesser–Pokrovskiy).

▷ Every odd-order group of order not divisible by 3 is R-harmonious: both vertices and cyclic products enumerate $G \setminus \{e\}$ (Javaheri, December 2025).

Can the cyclic ordering also respect inversion?

Symmetric Harmoniousness: A Complete Characterisation

Definition 37 (Javaheri, Def. 1.1)

A harmonious ordering $g_0, \ldots, g_{\ell-1}$ is symmetric if $g_{\ell-i} = g_i^{-1}$ for $1 \leq i \leq \ell - 1$.

Theorem 38 (Javaheri, AJC 2026, Thm. 1.2)

A finite group is symmetric harmonious if and only if it has odd order.

Previously, every odd-order group had some harmonious ordering. The 2026 publication guarantees one compatible with inversion. Preprint: November 2025; journal publication: 2026.

Symmetry in a Product-Coloured Cycle

Example 39 ($\mathbb{Z}_5$; addition in place of multiplication)

Vertex order: $0, 1, 2, 3, 4$. Cyclic sums: $1, 3, 0, 2, 4$. Opposite positions are paired by inversion: $4 = -1$, $3 = -2$.

In the complete digraph, colour $x \to y$ by xy.

▷ Listing every group element once makes the cycle Hamiltonian.

▷ Listing every adjacent product once makes it rainbow.

▷ $g_{\ell - i} = g_i^{-1}$ pairs positions in the vertex order by inversion.

This is the symmetric harmonious ordering in Javaheri's theorem. The symmetry is a condition on the ordering; no extra graph automorphism is asserted. The one-element group uses the algebraic definition.

Omitting the Identity from Both Lists

A group is **#-harmonious** if a cyclic ordering of $G \setminus \{e\}$ has cyclic products enumerating $G \setminus \{e\}$ (Def. 1.4). This is the same property called **R-harmonious** in Javaheri's December 2025 preprint.

In the product-coloured digraph, restrict the vertices to $G \setminus \{e\}$. The desired cycle visits every remaining vertex and uses exactly the nonidentity colours. It has length $|G| - 1$.

This differs from deck 02's R-sequencing: here an arc $x \to y$ has colour xy; there it has colour $x^{-1}y$.

New Non-Abelian Harmonious Families

Theorem 40 (Javaheri, 2026, Thm. 1.5)

Let K be abelian, neither a group with a unique involution, nor an elementary abelian 2-group, nor $\mathbb{Z}_3$. For odd-order groups $H_1, \ldots, H_k$, the iterated semidirect product

$$((\cdots(K \ltimes H_1) \ltimes H_2)\cdots) \ltimes H_k$$

is $\#$-harmonious. Here H_i is normal at the ith extension step.

Thus each such extension admits the rainbow cycle from the previous slide. The same symmetric-ordering theorem supplies the R*-sequenceability direct-product extension in deck 02.

Buratti–Horak–Rosa: Lists, Not Subsets

▷ Two lengths (Horak–Rosa, 2009); many three- and four-length lists and monotone multiplicity families (Ollis et al., 2020).

▷ (Ağırseven–Ollis, 2024) grid-based constructions extend the available three-length families.

▷ (Ağırseven–Ollis, December 2024; revised January 2025) isolate the coprime case: each length x satisfies $\gcd(x, v) = 1$.

▷ (Ağırseven–Ollis, 2025) further construction techniques for lists with small support.

A list L has $v - 1$ entries **with multiplicity**. A valid ordering theorem for a subset of a group does not automatically realise such a list.

Linear Realisations: The Translation and Its Limit

A linear realisation of L is a permutation $x_0, \ldots, x_{v-1}$ of $\{0, \ldots, v-1\}$ with

$$\{|x_i - x_{i-1}| : 1 \le i \le v-1\} = L \quad \text{as multisets.}$$

▷ Choose signed increments $d_i = x_i - x_{i-1}$: the initial x_0 and all partial sums must visit each label exactly once.

▷ Cyclic length is $\min(|x-y|, v-|x-y|)$. A linear realisation gives the desired cyclic list when every length is at most $\lfloor v/2 \rfloor$.

▷ Multiplication by a unit modulo v relabels the cyclic problem; a coprime support length can be normalised to 1 (§1).

Prescribed repeated edge lengths require more than distinct partial sums.

Recent BHR Families: Keep the Hypotheses

Write $L = \{1^a, x^b, y^c\}$ and $v = a + b + c + 1$.

Theorem 41 (Ağırseven–Ollis, 2024, a case of Thm. 1.3)

Suppose $1 < x < y \leq \lfloor v/2 \rfloor$, x is even, $a + b \geq y - 1$, and $a \geq x + y - 2$. Then L has a linear realisation, hence a cyclic realisation.

Theorem 42 (Ağırseven–Ollis, 2025, Thm. 1.9)

For $k \in \{1, 2\}$, $y > k + 1$ and $a \geq y$, the list $\{1^a, (y - k)^b, y^c\}$ has a standard linear realisation (one starting at 0).

For the second family, the cyclic interpretation again requires $y \leq \lfloor (a + b + c + 1)/2 \rfloor$. These are construction families; the full divisor-condition conjecture remains open.

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6. Open Problems

Open Problem 1: The Graceful Tree Conjecture

Conjecture 7 (Rosa, 1966/67)

Every tree T with n edges has an injective labelling $f\colon V(T) \to \{0, \dots, n\}$ whose edge differences $|f(x) - f(y)|$ are distinct.

⇕ Proposition 22

Graph form

For every tree T with n edges, the complete graph K_{n+1} on $\{0, \dots, n\}$, with xy coloured $|x - y|$, contains a **rainbow spanning copy of T**.

Known. caterpillars and many special classes; all trees with ≤ 35 vertices (Fang); $(1 - \varepsilon)n$ distinct differences for large trees (LPW, revised 2026); an approximate version for $\Delta = O(n/\log n)$ (AAGH). The conjecture "in full generality … remains open" (LPW, §1).

Open Problem 2: Ringel–Kotzig for Every n

Conjecture 8 (Kotzig; Ringel, 1963)

Every tree T with n edges has a ρ-labelling: an injective $f\colon V(T) \to \mathbb{Z}_{2n+1}$ whose edges have distinct lengths $\ell(f(x), f(y))$.

$\Updownarrow$ Proposition 21

Graph form

For every tree T with n edges, the ND-coloured K_{2n+1} contains a **rainbow copy of T**; equivalently, K_{2n+1} has a **cyclic decomposition** into copies of T.

Known. all sufficiently large n (MPS), (Keevash–Staden); every graceful tree, hence all trees with ≤ 35 vertices.

Open: the trees with 36 to n_0 vertices that are not known to be graceful; n_0 is not explicit.

Open Problem 3: The Graham–Sloane Conjecture

Conjecture 9 (Graham–Sloane, 1980)

Every tree T with n edges has a labelling $f\colon V(T) \to \mathbb{Z}_n$, with exactly one value used twice, whose edge sums $f(x) + f(y)$ are distinct.

$\Uparrow$ Proposition 23: the graph form is stronger

Graph form (Müyesser–Pokrovskiy, Conj. 1.3)

For every tree T on n vertices, the sum-coloured $K_{\mathbb{Z}_n}$ contains a **rainbow copy of** T.
(Chang–Hsu–Rogers, 1987: even one avoiding the colour 0.)

Known. all trees with ≤ 31 vertices (Fang); caterpillars; bounded-degree trees for large n (Müyesser–Pokrovskiy, 2025).
Open: trees of unbounded degree.

Open Problem 4: Rainbow Trees in All Sum Colourings

after Müyesser–Pokrovskiy, §1

For which pairs (T, G) — T a tree on n vertices, G an abelian group of order n — does K_G contain a rainbow copy of T?



definition of K_G

Labelling form

For which (T, G) is there a bijection $f\colon V(T) \to G$ whose edge sums $f(x) + f(y)$ are distinct?

Known. the answer for bounded-degree trees and large n: exactly when none of the three obstructions occurs (Thm 1.5).
Open: unbounded degree. "We were unable to come up with a plausible conjecture of a characterisation" (Müyesser–Pokrovskiy, §1).

Open Problem 5: Which Groups Are Harmonious?

expected; see Javaheri, §1

Is every finite group that satisfies the Hall–Paige condition, other than $\mathbb{Z}_2^k$, harmonious: can its elements be listed $g_1, \ldots, g_n$ with $g_1 g_2, \ldots, g_n g_1$ all distinct?

$\Updownarrow$ Proposition 24

Graph form

For every such G, does the complete digraph on G, with $x \to y$ coloured xy, contain a **rainbow directed Hamilton cycle**?

Known. the eligible abelian groups; odd order; dihedral and dicyclic of order $8n$ ($n > 1$); sufficiently large eligible groups (MP). Odd-order groups even admit symmetric harmonious orderings (Javaheri, 2026). The remaining even-order cases are open.

Open Problem 6: The Buratti–Horak–Rosa Conjecture

Conjecture 10 (Horak–Rosa, 2009; partial-sums form)

If L is a multiset of $v-1$ elements of $\{1,\dots,\lfloor v/2\rfloor\}$ and every divisor d of v divides at most $v-d$ of them, then L can be ordered and signed so that the partial sums of $\pm a_1,\dots,\pm a_{v-1}$ are exactly $\mathbb{Z}_v\setminus\{0\}$.

⇕ Proposition 25

Graph form

Under the same condition, K_v has a **Hamilton path whose multiset of edge lengths is L**; equivalently, the ND colouring has a Hamilton path with colour multiset L.

Known. two lengths; $v\le 18$ and primes $p\le 23$; three- and four-length families (OPPS); grid-based and small-support families (2024), (2025). The full divisor-condition conjecture remains open in the sources checked.

Summary

▷ **Ringel** (1963) asked to decompose K_{2n+1} into copies of any tree; **Rosa** turned it into a labelling problem; **Kotzig** into a cyclic one.

▷ **Graham–Sloane** (1980) came from additive bases and codes; their harmonious labellings led **Beals et al.** (1991) to harmonious groups.

▷ **Hahn**, **Buratti**: Hamilton paths with constrained colours; the group answers or obstructs.

▷ **The translation**: a labelling by $\mathbb{Z}_m$ is a copy of the tree in K_m; distinct differences / sums = **rainbow** in the ND / sum colouring.

Differences colour by length; sums colour by direction.

Thanks !

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